

On the Economic Foundation of the Urban Network Paradigm: Spatial Integration, Functional Integration and Economic Complementarities within the Dutch Randstad

Frank van Oort, Martijn Burger and Otto Raspe

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Abstract

Conceptually, the degrees of spatial and functional integration and urban complementarities in economic network relations are hypothesised to be important. In this paper, data on interfirm relations in the Dutch Randstad are used to test conditions for integration and the existence of economic complementarities. A clear hierarchy is observed in the different types of spatial interdependencies in the Randstad, in which the central place model prevails. Furthermore, no evidence is found for the functional integration of municipalities in the Randstad. It is concluded that, at this moment, the Randstad does not function as a spatially and functionally integrated region and that spatial economic policy can better focus on smaller regions within the Randstad. This also calls into question the applicability of the urban network concept in general, as the Dutch Randstad is usually seen as a prime example of an economically successful polycentric urban system.

1. Introduction

From a geographical point of view, the Randstad is known as the urban conurbation in the western part of the Netherlands, in which four major cities (Amsterdam, Rotterdam, The Hague and Utrecht) and a number of smaller towns are located in an area of approximately 90 by 90 km (see Figure 1). The Randstad constitutes the heart of the Dutch economy, with 50 per cent of the gross national product being generated on

Frank van Oort is in the Department of Economic Geography, Utrecht University, P.O. Box 80115, Utrecht, NL-3508 TC, The Netherlands. E-mail: f.vanoort@geo.uu.nl.

Martijn Burger is in the Department of Applied Economics, Erasmus University Rotterdam, PO Box 1738, Rotterdam, NL-3000 DR, The Netherlands. E-mail: mburger@ese.eur.nl.

Otto Raspe is in the Netherlands Environmental Assessment Agency (PBL), PO Box 30314, The Hague, NL-2500 GH, The Netherlands. E-mail: otto.raspe@pbl.nl.

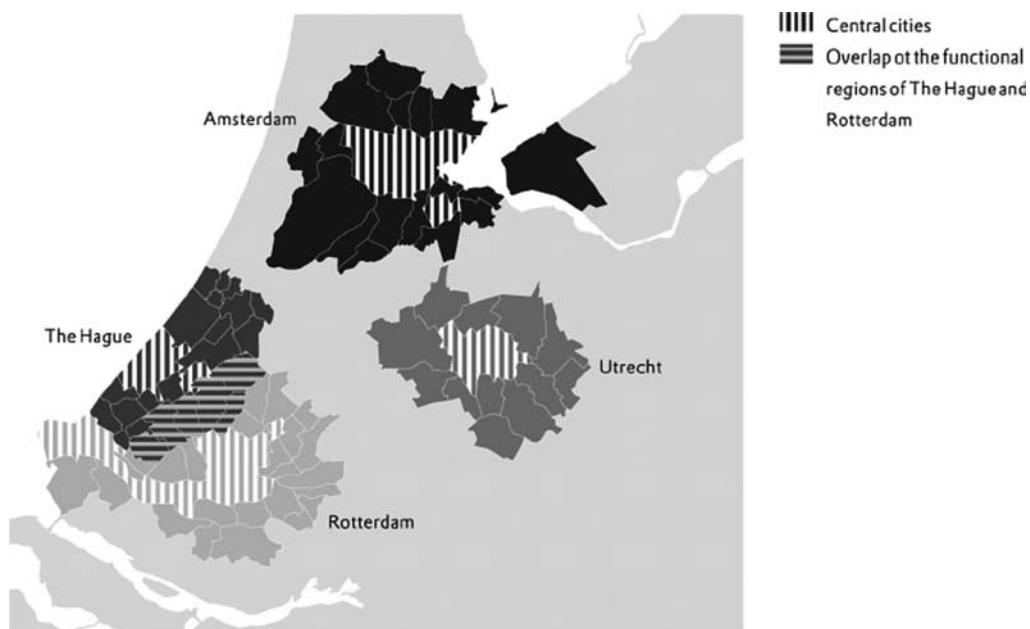


Figure 1. The Randstad research area (90 x 90 km) and the 67 municipalities in the survey.

approximately 25 per cent of the country's total land area. Having a population of six million inhabitants, the Randstad houses over one-third of the Dutch population. However, in the debate on spatial planning and economic policy in the Netherlands, the 'Randstad' stands for more. The name, since it includes the Dutch word for 'city' (*stad*, in the singular), suggests that the Randstad is a single, contiguous urban region. Suggestions that the region functions as an integrated economic entity for basic industries, like manufacturing activities, distribution activities and business services, are numerous (see for example, Dieleman and Musterd, 1992; Sachar, 1994; Batten, 1995; Kloosterman and Lambregts, 2001; Lambregts, 2008). Based on these suggestions, policy-makers now more than ever aim at the concentrated location of (inter)national firms and businesses in this networked region in order to have optimal economic growth potentials (VROM, 2008). Nevertheless, simply assigning a name, such as the Randstad, to a collection of towns and cities

does not automatically meld them into a spatial and functional integrated city with economic complementarities and firms benefiting from region-wide agglomeration economies.

From the 1960s onwards, the Randstad has been regarded internationally as one of the most important urban networks or functional clusters of economic activity in Europe (Hall, 1966; Batten, 1995). It is argued that the Randstad functions as a role model for a network of cities and towns which, given their mutually dependent specialisations and their variety, creates a favourable setting for economic production and growth (Lambregts *et al.*, 2006). In the urban geography and planning literature, it is generally contended that such cities and towns complement each other in terms of economic specialisations, each being therefore more competitive than they would be in isolation (Meijers, 2005). This representation of the Randstad as urban network is based on the assumption of a considerable regional cohesion

in personal, occupational and corporate relationships of people, organisations and firms that transcends the boundaries of the traditional metropolitan areas. The general contention for different cities within an urban network to be complementary to each other requires that the cities not only be specialised in different industries but at the same time also be showing a marked degree of spatial interaction and hence integration. In this, interfirm relationships are considered to be most fundamental to the creation and evolution of such an urban network (Sachar, 1994; Lambooy, 1998). Camagni and Capello (2004, p. 496) argue that polycentric urban regions comprise a new paradigm in spatial sciences, provided that: the exact meaning is defined; the theoretical economic rationale is justified; and the novel features of its empirical content are clearly identified and distinguished from more traditional spatial facts and processes that can be interpreted through existing spatial paradigms. Camagni and Capello further argue that

the concept of spatial networks is sometimes merely used as a substitute for 'interaction': an exchange of goods, services, information and contacts among places and nodes. Traditional paradigms of spatial interaction can be easily utilised, unless one could demonstrate that the probability of such exchanges is mainly independent of distance and the size of nodes (Camagni and Capello, 2004, p. 496).

This paper contributes to the discussion on economic complementarities in urban networks by analysing networked economic relations in the Randstad. To date, there has been little empirical research into the spatial and functional economic cohesion of urban-networked regions, mainly because of a lack of interfirm relational data. We use a recently collected dataset that stems from a large survey of 1676 firm establishments that are active in the industry, producer services and distribution sectors in the

Randstad to test for the presence of urban complementarities. A detailed insight into interfirm networks is obtained by asking these firms to identify their most important business relationships with customers and suppliers in terms of physical goods, services and information. Using these data, which are representative of the entire commercial sector in the Randstad, and defining urban complementarities by the combination of spatial and functional integration between localised specialised firms, we answer three related questions on the economic foundation of urban networks

- (1) To what extent does the Randstad operate as a spatially integrated entity?
- (2) To what extent does the Randstad operate as a functionally integrated entity?
- (3) Do urban complementarities between municipalities (being the joint occurrence of spatial and functional integration) exist in the Randstad?

The rest of this article is organised as follows. Section 2 discusses the literature on the spatial and functional development of cities and urban systems, focusing primarily on the role of economic networks in the concepts of spatial integration, functional integration and economic complementarities. Section 3 offers a diagrammatic representation of the network relationships of the surveyed firms. Section 4 examines the degree of spatial integration of interfirm relationships in the Randstad by means of spatial interaction models. In section 5, we inspect the degree of functional integration by looking at the degrees of specialisation and diversification across municipalities in the Randstad in relation to the interactions between firms in these municipalities. Finally, in section 6, we examine the presence of urban complementarities in the Randstad, conclude on our three research questions and sketch out policy implications.

2. The Economic Foundation of Changing Urban Systems

2.1 The Monocentric Model: Towns into Cities and Daily Urban Systems

In order to judge whether the Randstad can be classified as a single urban economic entity, a classification of urban functions in relation to the morphology of cities is necessary. Prior to the Industrial Revolution, a city could be identified as a monocentric agglomeration of people and businesses (a town), clearly separated from the countryside. What made cities most distinctive from the rural areas was their specific legal status (notably 'city rights') and their political identity. Markets and administrative places gradually

crystallized as central places for their surrounding areas in which each center approximated the center of gravity of its service area in order to minimize transportation costs (Hohenberg and Lees, 1985, p. 49).

From a morphological point of view, the city in the pre-industrial era was then also very concentrated and centralised, as physical distance often encumbered interactions between different areas. As a result, business relations were often restricted to the urban core.

Cities shifted their borders when mobility steadily increased and communication technology further developed. The tram, train and telegraph in the 19th century and the telephones and motor vehicles in the 20th century are examples of technology that impacted urban development (Anas *et al.*, 1998). Also, due to the existence of cheaper land outside the urban centre and rising welfare levels, households increasingly choose to locate further away from the city centre (Pred, 1973). These developments led to the rise of metropolitan areas or daily urban systems (DUS), which are typically defined by the most important commuting relations (van der Laan, 1998). The relationship between the urban core and its suburbs remained

hierarchical-nodal. Most economic activity was based in the urban core, commuting flows were directed towards the central cities and suburbs only fulfilled a residential function. Here, the metropolitan region is characterised as a central place system with the central city as main focus. In line with this well-defined morphological and functional development of cities, the traditional starting-point for a treatment of the theory on urban systems then was Burgess' concept of the monocentric city (Burgess, 1925), later formalised in a location and land use theory by Alonso (1964). The concept of the monocentric city involves a central business district, surrounded by a circular residential area whereby land is allocated according to its most profitable use. The general idea of the monocentric city is that most economic activities are based in the urban core, whereas suburbs fulfil a residential function. Hence, the relationship between the urban core and its suburbs in the monocentric model is hierarchical-nodal or centralised in the sense that commuting flows are predominantly directed from the suburban areas towards the central cities. The monocentric theory of urban development further suggested that interfirm business flows largely remain within or between urban cores (Pred, 1973, 1977), although space-consuming economic activities—like industrial and distribution activities—are hypothesised to develop radially out of the city centre (Smith, 1971; Phelps and Ozawa, 2003). From a morphological point of view, this radial evolution of population and employment, causing the transformation of towns into cities into regionalised daily urban systems, is characterised as a shift in spatial scale (Kloosterman and Musterd, 2001).

2.2 The Polycentric Urban Region (PUR)

The focus of the contemporary debate on changing urban systems has increasingly shifted from the intraurban scale to the inter-urban scale (Kloosterman and Musterd, 2001;

Hall and Pain, 2006; Hoyler *et al.*, 2008; Bailey and Turok, 2001; Coe and Townsend, 1998). The interurban scale corresponds to flows *between* different urban regions. Usually, urban systems are blended in terms of intraurban and interurban relations, in which intraurban relations refer to relations in which the 'own' core and periphery districts are involved and interurban relations refer to relations between core and peripheral districts in different urban regions. As catchment areas of different cities start to overlap, individual metropolitan areas lose significance as independently functioning daily urban systems and could instead be perceived as forming parts of an urban network. Much of the current literature is focused on this development—that is, the development of the polycentric urban region (PUR). The PUR can be represented as an urban network of historically and spatially separate metropolitan areas comprising a larger region (Bourne and Simmons, 1978; Meijers, 2007). These metropolitan areas can be network-cities themselves, but this is not necessarily the case (i.e. the urban system can be dominated by a polycentric structure at the interurban level and a monocentric structure at the intraurban level or vice versa). Likewise, urban network formation at the interurban level is not necessarily the next evolutionary step after the (intraurban) network-city (Parr, 2004; Graham and Marvin, 2001).

Social and economic processes are taking place in ever-larger geographical areas, in which suburban districts increasingly emerge into local centres that develop their own economic activities and because of this start competing with the original urban core (Garreau, 1988). The result is the development of cities with multiple centres, or polycentric cities. In such a polycentric city, commuting is no longer centralised but is instead reciprocal in the sense that flows are directed not only from the suburbs to the urban core but also from the urban core to the suburbs. The spatial range of functional relations corresponds to commuting and

firm-relational flows that remain within the urban region. Flows of goods, services and people become decentralised as the number of workers commuting between suburbs and bypassing the old urban core increases (criss-cross relations). More specifically, economic complementarities within the daily urban system (DUS) arise, where one location may be regarded as 'central' in terms of one particular function while other places might be central in terms of other functions. This coincides conceptually with the identification of the paradigm shift suggested by Camagni and Capello (2004): from a central place model to an urban network model. The former is characterised by periphery–core relations (for commuting, shopping and lower-order interfirm relations) and inter-core relations (for higher-order interfirm relations), while significant periphery–core, criss-cross and intraurban interfirm relations should characterise the latter.

2.3 The Economic Foundations of Urban Networks

The conceptualisation of urban network formation is predominantly tested by interaction patterns of people. Previous research has focused on the effect of the increasing flexibility and mobility of people and their changing residential preferences on the urban system (Renkow and Hoover, 2000; van Ham, 2002). To some extent, these residential preferences are mutually influenced by enhanced mobility, the increasingly flexible workplace and choices of lifestyle and attitudes of households, including the locational preferences of two-earner households, the increasing number of working women and an increasing number of single-person households (Hall, 2001). Still, these trends in labour supply do not, on average, coincide with an increasing labour mobility of employees: the average commuting time in the Netherlands has remained stable over the past 20 years at approximately 25 minutes (van der Burg and

Dieleman, 2004). Only a limited number of higher-educated professionals have become more mobile over time.¹

Instead, *firms* are argued to have become the most mobile and flexible over recent years. This would imply that urban networks form an enlarged home base for firms in terms of market potential, diversity and quality of available knowledge, infrastructure, institutions and sub-contracting possibilities (Pred, 1977). This microeconomic (business) foundation of urban networks is hinted at in the literature (Kloosterman and Lambregts, 2001; Lambregts *et al.*, 2006), but has not been much researched in the urban geography and planning literature. This is not surprising, since urban economists have for some time fuelled a burgeoning empirical literature that examines whether spatial circumstances give rise to agglomeration economies—external economies from which firms can benefit through co-location and local network creation—that endogenously induce localised economic growth (Glaeser *et al.*, 1992; Combes, 2000; Rosenthal and Strange, 2003). In their survey of the empirical literature on the benefits of agglomeration, Rosenthal and Strange (2004) point out that the elasticity of productivity to city and industry size typically ranges between 3 per cent and 8 per cent. However, the effects of agglomeration economies on localised economic growth differ across sectors, time and space (Rosenthal and Strange, 2004; van Oort, 2007). Concerning the latter, there is a growing interest in this debate on agglomeration economies in the networked structure of (systems of) cities (Anas *et al.*, 1998; Hall and Pain, 2006).

At the micro level, the relation between the morphology of cities or city-systems and economic performance is conceptualised by the increasing flexibility and mobility of firms. The hypothesis is that trends in urban system dynamics are driven by changes in the spatial distribution of employment and growth opportunities (Renkow and Hoover, 2000).

Aoyama and Castells (2002) argue that these changes are facilitated by key advancements in transport technology and in information and communication technology (ICT), by globalisation and flexible specialisation in production and by the economic change of most Western economies from a production to a services-dominated economy. There is considerable academic debate on the precise spatial and economic effects of these developments. Concerning ICT, for instance, in one view—the substitution view—it potentially influences the production of new products and the location of their production by substituting for production factors such as manual labour or traditional means of transport. This then leads to higher efficiency and productivity in firms. In substituting physical transport by ICT—mainly relevant for services—a reduction in time and transaction costs can be achieved. A liberalising or centrifugal effect then occurs, in the literature summarised as the ‘death of distance’ or the ‘weightless economy’, in which economic value is transmitted across physical space at zero marginal costs (Horan *et al.*, 1996; Kolko, 2002). The so-called complementary view on the other hand stresses inertial and concentrated growth patterns. In this conceptualisation, spatial economic dynamics occur within the limits of the spatial interaction patterns laid down in the past in processes characterised by cumulative causation. Moreover, firms are also physically constrained by the necessity of face-to-face contact (Geyer, 2002). Functional and trust relations thus limit the morphological layout of urban agglomeration to that of the (polycentric) region (Nooteboom, 1999). Comparable discussions concern the (spatial) flexibility thesis of firms. It is argued that, to compete in the network economy, firms have to make production processes more flexible with respect to time, place, contracts and job content (Scott, 1988). Increasing uncertainty in markets and differentiation in consumer demand in terms of varieties, brands and

quality force firms to run small production batches. The outsourcing of economic activities that do not directly belong to the core activities of the company is stimulated and vertical co-operation with firms active within the same industry becomes more important. This flexibilisation of business processes and the functional division of tasks between companies create opportunities for a spatial division of labour and different spatial settings or locations become suitable for different economic functions. The result is often argued to be a polycentral and multinodal structure, in which flows of goods, services and people are not one-sided, but two-sided and criss-crossed (Curran and Blackburn, 1994).

2.4 Integration and Complementarities in Local and Regional Policy

Local and regional policies often have the explicit intention of economic deconcentration and urban network formation. A well-known example of an intentional policy towards economic deconcentration is the 'growth pole concept' (Parr, 1999). The aim of this policy is to create economic development in peripheral areas by encouraging the establishment of industrial growth centres in the periphery. Ultimately, this should have led to the development of the hinterlands of these growth centres, in turn spreading the benefits of economic development over a larger area. Similarly, by embracing the urban network concept, national as well as local policy-makers and urban planners attempt actively to develop suburban and adjacent areas, with the objective of spreading economic prosperity and enhancing territorial competitiveness of urban regions (Glaeser, 2007). In the Netherlands also, the urban network paradigm is embraced by policy in this vein. In recent policy memoranda by the Ministry of Economic Affairs (*Peaks in the Delta*; EZ, 2005), the Ministry of Housing, Spatial Planning and the Environment (*Spatial Memorandum*; VROM,

2004; and *Structural vision Randstad 2040*, VROM, 2008), and the Ministry of Transport (*Mobility Memorandum*, V&W, 2004), the concept takes in a prominent place. Three key concepts within the present-day academic as well as policy conceptualisation of urban networks are spatial integration, functional integration and urban complementarities. Spatial integration between the towns in an urban network (for instance, in the Randstad) results from the improved opportunities for mobility and communication, which facilitate larger transport and information flows and stronger relationships between the individual urban agglomerations. An urban network can be classified as a functionally integrated entity if alongside more spatial integration there is differentiation between the municipalities in terms of urban economic functions. In an economic sense, this would require that the municipalities that make up an urban network (the Randstad) be specialised in different economic sectors, thereby fulfilling different economic roles (Meijers, 2005). For example, a city specialised in business services provides these services to a city specialised in labour-intensive industry and vice versa (Lambregts *et al.*, 2006). In sum, differences in relative specialisations (functional integration) together with a large degree of interaction between economic agents (spatial integration) economically define an urban network that is characterised by economic complementarities.

In earlier days, integration and complementarities bundled in one town or city, but with the increasing flexibility and mobility of firms, their goods and services and the division of labour, it is suggested that these processes are now embedded in regionally-defined networks. Despite the fact that both the academic literature and policy documents on urban networks use the concepts of spatial and functional integration and economic complementarities, only few empirical studies have qualitatively assessed how well the

Table 1. Number of establishments, by region and sector

	<i>Population</i>	<i>Sample</i>	<i>Responses (n)</i>	<i>Response rate (percentage)</i>
<i>Region</i>				
Amsterdam	19 045	7 035	574	8.2
Rotterdam	10 789	5 668	514	9.1
Den Haag	5 468	3 655	291	8.0
Utrecht	6 096	3 943	297	7.5
Total	41 398	20 301	1 676	8.3
<i>Sector</i>				
Manufacturing	5 322	5 307	367	6.9
Wholesale	10 991	4 807	376	7.8
Business services	25 085	10 186	933	9.2
Total	41 398	20 301	1 676	8.3

network model fits the reality of contemporary urban systems (Davoudi, 2003; Capello, 2000). Unfortunately, most of the available empirical evidence is based on node characteristics, by which (interfirm) interaction patterns can be mainly explained by distance and the size of nodes, using methods such as location quotients, rank–size relations, sufficiency indices and employment-to-work ratios (Limtanakool *et al.*, 2007; Camagni and Capello, 2004). In our empirical analysis, we will introduce a method based on flow characteristics that tests for the spatial structure of interfirm relations independently of masses of nodes and the distances between them.

3. Interfirm Relations in the Randstad

3.1 Data: A Survey among Firms in the Randstad

In the previous section, we defined urban complementarities as the *combination* of spatial integration and functional integration. To test whether the Randstad does indeed function as such a single economic system or urban network with complementary sub-economies, we have examined the spatial dynamics of firms' networks within the region. The network relationships involve (sub-)contracting,

purchasing and selling of products, services and knowledge. In our analyses, we aggregate the relationship flows into municipalities (LAU-2) of which there are 69 in the Randstad (see Figure 1). The survey was conducted in 2005 among more than 20 000 selected firms (establishments) with more than one employee and based in the Dutch Randstad (divided in four sub-regions, see Figure 1). A random stratified sample, taking size, sector and regions into account, was taken from the LISA database (an employment register of all Dutch economic establishments; see van Oort, 2007). For this research, we included firms in manufacturing, wholesale and business services. These basic sectors are not directly bound to consumers for their location—as retail is, for example. The response of 1676 establishments—approximately 8 per cent—is representative of the stratification by region, size and sector. The questionnaire focuses on the sources and destinations of an individual firm's 10 most important selling and purchasing relations, which we aggregate to the municipality level in our analysis. Table 1 shows for each of the four sub-regions in the Randstad (Amsterdam, Rotterdam, The Hague and Utrecht) the firm population, the firm sample size and the number of firms responding to the questionnaire. The calculation of

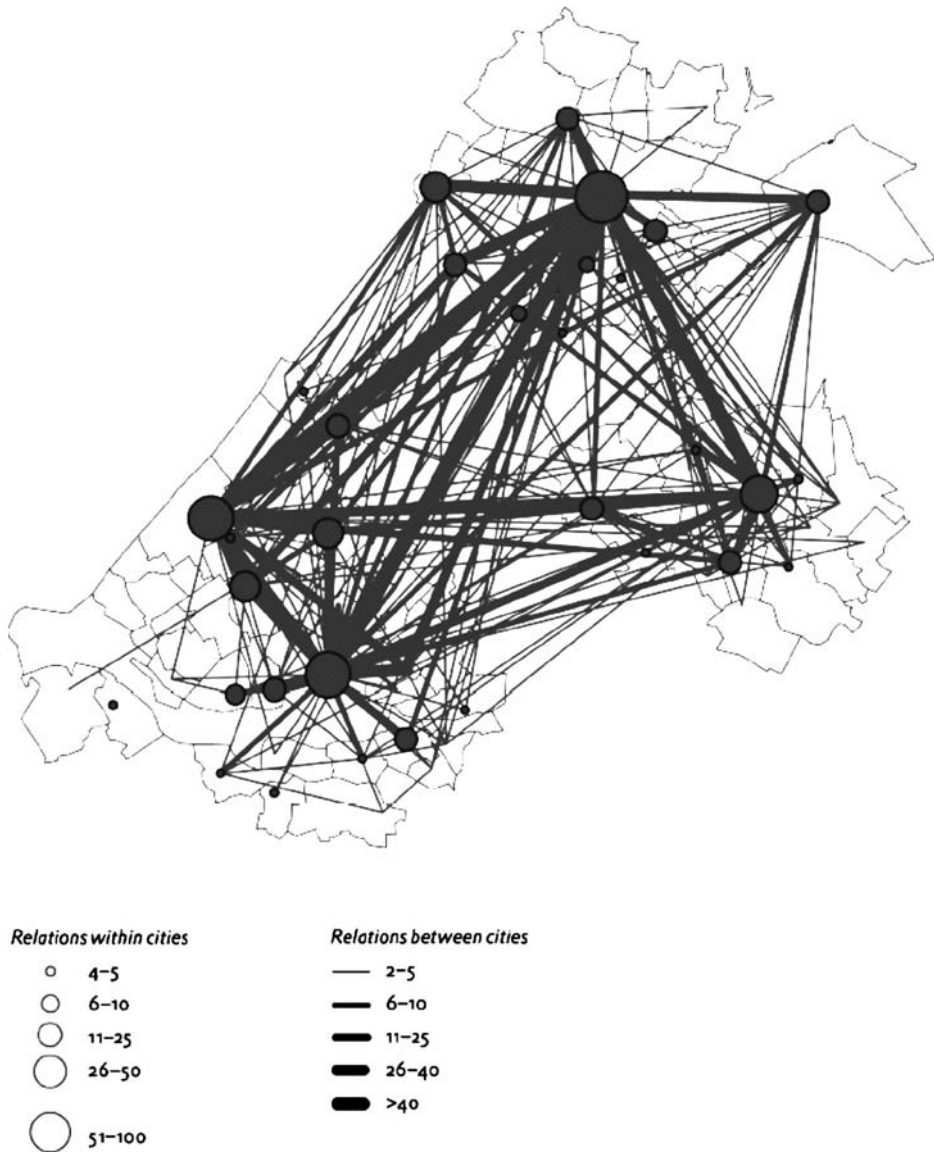


Figure 2. The network of interfirm relationships in the Randstad.

the sample size is based on stratification by region, economic activities (by sector, 27 NACE codes in manufacturing, wholesale and business services) and firm size (six size classes).

3.2 Visualisation of Network Patterns within the Randstad

Figure 2 shows the network of interfirm relationships for the entire Randstad. In

this figure, the respondent population is not classified by sector or size. The dots indicate the extent of relationships within an individual town or city—for example, companies in The Hague that maintain a business relationship with other companies in The Hague. The lines represent the relationships between companies in different towns or cities. A large number of relationships can be

seen within the four major cities, but there are also flows between these main centres. Moreover, a significant number of criss-crossing relationships—i.e. relationships between companies that are located outside the central cities—can be noticed. In absolute terms, the four major cities function as the centres of the urban network. Not only do companies there maintain the largest number of relationships with other firms in the same municipality, but they also are involved in a large number of flows within the larger network. This is particularly true in the case of Amsterdam, which occupies a key position in the Randstad's overall network. It should be noted, however, that Figure 2 is only telling half of the story: about 50 per cent of all interfirm relationships of firms that are located in the Randstad region transcend the boundaries of the Randstad.

Figure 2 gives a descriptive account of the embeddedness of economic networks in the Randstad region. However, an obvious criticism of such a visual analysis is that it does not allow for differences in the absolute sizes of municipalities (in terms of firm population) and the physical distances between them. For example, recent empirical studies conducted within the POLYNET research framework (for example, Green, 2007; Hall and Pain, 2006) do not address this issue. The likelihood of an interfirm relation with companies located in a large city with many (and a sectorally varied number of) firms is larger than one directed to a smaller city with fewer firms. Likewise, the likelihood of an interfirm relation between two cities in close proximity to each other is larger than the likelihood of one between municipalities located far from each other. Figure 3 therefore shows the same map as Figure 2, but the visualised spatial interactions are now characterised by their deviation from the number of expected relationships based on the size of the origin and destination municipalities and their mutual physical proximity. In order to compare the observed

versus the expected interaction intensity between cities in the Randstad region, we made use of the likelihood ratio chi-squared statistic (Agresti, 2002). Where the number of actual relationships is significantly larger than the expected figure, this is shown in dark gray shading. Where the number of actual relationships is significantly smaller than the expected figure, this is shown in light gray shading. Non-significant relations are shown in Black. Figure 3 shows that interfirm relationships within the same municipality and relationships between central cities and their direct (adjacent) hinterland are in general larger in number than projected. Interfirm relations between the four central cities are less than expected.

All locations and regions in the modern economy interact to some extent (Fingleton, 2003). Accordingly, based on Figure 2 we cannot draw conclusions on the nature of the existing urban interdependencies within the Randstad region. And, although Figure 3 suggests a spatial hierarchy in more-than-expected interactions, it is not precise in the degree of this ranking and does not inform us on functional interactions. We therefore introduce an interaction model that controls for the masses of cities and their mutual physical proximity to analyse spatial and functional integration more accurately.

4. The Spatial Economic Integration of the Randstad

4.1 Modelling Spatial Integration

In this section, we test whether the municipalities in the Randstad form a spatially integrated urban system based on interfirm relations. In this, we use a rather strict testing system in the sense that we speak of spatial integration when there is no effect of spatial-functional context on interfirm network intensities other than the economic mass of sending and receiving localities and the physical distance between them. If the Randstad

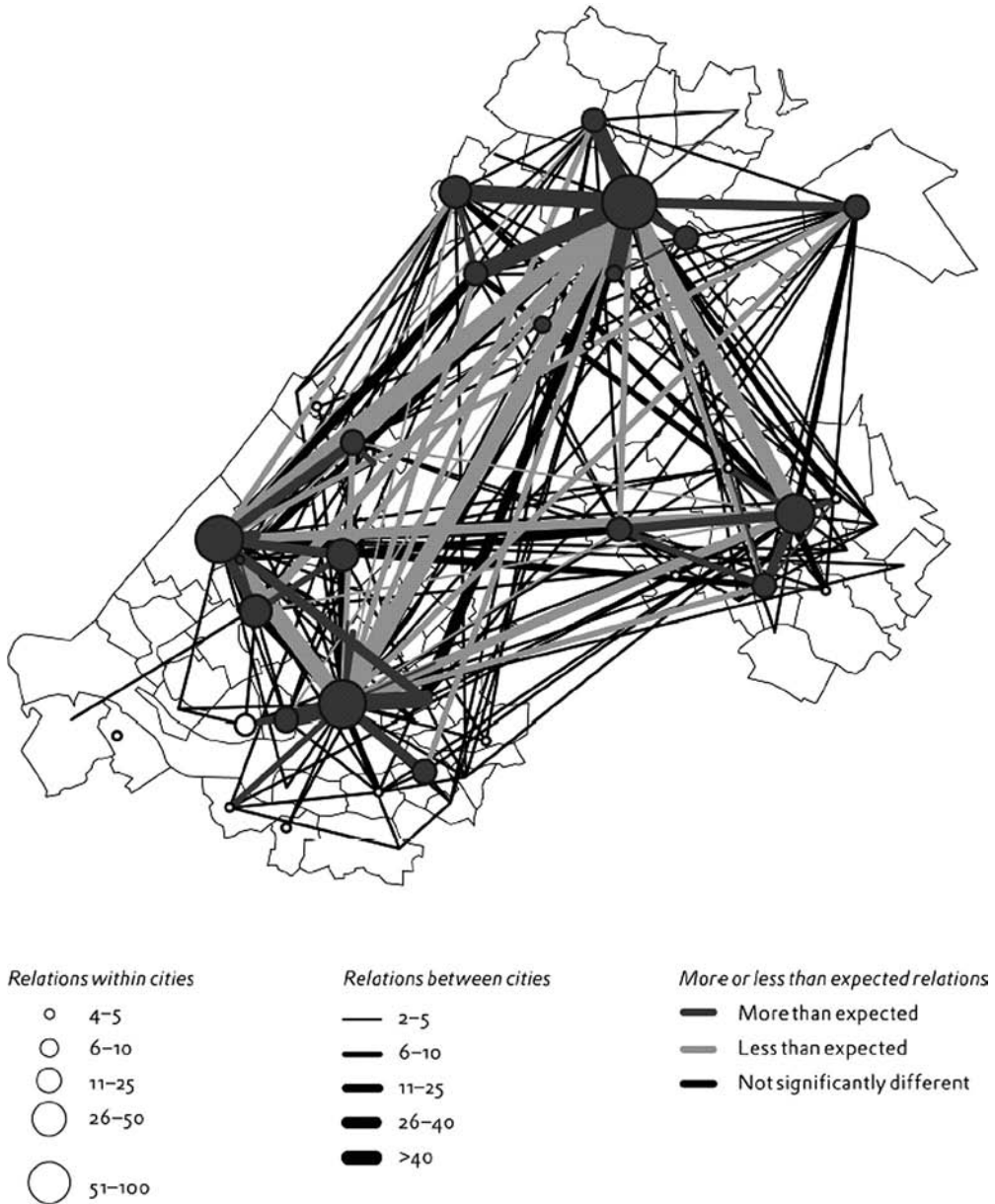


Figure 3. The network of inter-firm relationships in the Randstad, classified by whether the relationships are more or less than expected.

functions as a spatially integrated urban system (from an economic point of view), the spatial network structure of interfirm relations in the Randstad is solely determined by these two variables. Three related conditions for spatial integration are, therefore

- (1) Intraurban interdependency should not be stronger than interdependencies between cities within the Randstad.
- (2) Interdependencies between cities within one of the urban (sub-)regions in the Randstad should not be stronger than

interdependencies between cities across these (sub-)regions.

- (3) No observable hierarchy in the different types of intermunicipal interdependencies should be present.

We test these conditions within the context of a gravity model. Spatial interaction patterns, such as interfirm flows, can be predicted and elucidated in analogy with Newton's law of universal gravitation (Haynes and Fotheringham, 1985; Sen and Smith, 1984). The gravity model assumes that the gravitational force between two objects (in our study, the interaction between companies, aggregated at the level of the municipality) is dependent on the mass of the objects and the physical distance between them. More specifically, the interaction intensity between two municipalities (the origination and destination) is hypothesised to be directly correlated with the masses of the municipalities and inversely correlated with the physical distance between the two municipalities. More formally

$$I_{ij} = K \frac{M_i^{\beta_1} M_j^{\beta_2}}{d_{ij}^{\beta_3}} \quad (1)$$

where I_{ij} is the gravitational force, or in our case, the interaction intensity between municipality i and j expressed by the number of linkages; K a proportionality constant; M_i the mass of the municipality of origin; M_j the mass of the municipality of destination; d_{ij} the physical distance between the two municipalities; and β_1 , β_2 and β_3 are parameters to be estimated.

Here, the masses of the municipality of destination and of origin are defined as the total number of interfirm relations of each municipality, as embedded in the network. As interfirm

relationships are considered to be undirected in this study, there is no clear distinction between the municipality of origin and the municipality of destination. Hence, the natural logarithm of the product of M_i and M_j is taken together as one variable in our analysis. Distance is measured as the crow flies between cities, where intramunicipal distances are calculated by

$$d_{ii} = \frac{2}{3} \sqrt{\frac{A_i}{\pi}} \quad (2)$$

in which the intramunicipal distance d_{ii} is two-thirds of the radius of the presumed circular area A_i (see Bröcker, 1989, and Frost and Spence, 1995, for the exact derivation and overview of considerations of this).

Spatial interaction data should be handled as count data, as they 'count' the number of times that something has happened, which in our case is the frequency of flow (expressed as interaction intensity) between and within municipalities. Although these data are often treated as though they are continuous, the application of the conventional linear regression model here can lead to inefficient, inconsistent and biased estimates (Flowerdew and Aitkin, 1982; Long, 1997) as the underlying assumptions of normal distribution and homoscedasticity are often not satisfied. For this reason, the use of alternative regression techniques is more appropriate. The most common regression model applied to count data is probably the Poisson regression, which is estimated by means of maximum likelihood estimation techniques. In this log-linear model, the observed interaction intensity between municipality i and j has a Poisson distribution with a conditional mean (μ) that is a function of the independent variables. More specifically

$$\Pr[I_{ij}] = \frac{\exp(-\mu_{ij}) \mu_{ij}^{I_{ij}}}{I_{ij}!} \quad (3)$$

where, $\mu_{ij} = \exp(K + \beta_1 \ln(M_i M_j) + \beta_2 \ln d_{ij})$

In order to correct for overdispersion (where conditional variance is larger than the conditional mean) and an excessive number of zero counts in our dataset (where the incidence of zero counts is greater than would be expected for the Poisson or negative binomial distribution), we make use of the zero-inflated negative binomial regression, which can be perceived as an extension of the Poisson model. Not correcting for the overdispersion and excess zero problem normally results in consistent but

inefficient estimates, exemplified by spuriously large z-values and spuriously small p-values due to downward-biased standard errors (Gourieroux *et al.*, 1984; Cameron and Trivedi, 1986). The zero-inflated negative binomial model considers the existence of two (latent) groups within the population: a group having strictly zero counts and a group having a non-zero probability of counts different from zero. Correspondingly, its estimation process consists of two parts (4.1 and 4.2)

$$\Pr[I_{ij} = 0] = \psi_{ij} + (1 - \psi_{ij}) \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_{ij}} \right)^{\alpha^{-1}} \quad (4.1)$$

$$\Pr[I_{ij}] = (1 - \psi_{ij}) \frac{\Gamma(I_{ij} + \alpha^{-1})}{I_{ij}! \Gamma(\alpha^{-1})} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_{ij}} \right)^{\alpha^{-1}} \left(\frac{\mu_{ij}}{\alpha^{-1} + \mu_{ij}} \right)^{I_{ij}} \quad (4.2)$$

in which, $\mu_{ij} = \exp(K + \beta_1 \ln(M_i M_j) + \beta_2 \ln d_{ij})$; ψ_{ij} is the proportion of observations with a strictly zero count ($0 \leq \psi_{ij} \leq 1$); $\Gamma(\cdot)$ is the gamma function; and α is a parameter that determines the degree of dispersion in predictions, thereby allowing the conditional variance to exceed the conditional mean.

The first component of the zero-inflated negative binomial model (4.1) contains a logit regression of the predictor variables on the probability that there is no interaction between two given municipalities at all. The second component (4.2) contains a negative binomial regression on the probability of each count for the group that has a non-zero probability of count different from zero. A likelihood ratio test of overdispersion (Cameron and Trivedi, 1986) is employed to test whether the negative binomial distribution is preferred to a Poisson distribution, while the Vuong statistic (Vuong, 1989) provides evidence of whether a zero-inflated model is favoured above its non-zero-inflated counterpart.²

In order to test the three specified conditions for spatial integration of municipalities in the Randstad region using a zero-inflated

negative binomial model, we introduce dummy variables in our model that reflect the spatial-functional context of the economic interactions between types of localities (regimes). These regimes build up the degree of spatial integration in the Randstad region. We distinguish four core cities (Amsterdam, Rotterdam, The Hague and Utrecht) with their suburban municipalities and introduce the following six types of urban interdependencies that convey the spatial context of flows between municipalities

A. Economic relations within municipalities

1. Intranodal relationships that remain entirely within a particular municipality (this can be either the central city or a suburban location).

B. Economic relations within urban (sub-) regions of the Randstad

2. Core-periphery interdependencies between the central cities and one of its 'own' suburban municipalities.
3. Interperiphery or criss-cross interdependencies between the suburbs within a given urban (sub-)region.

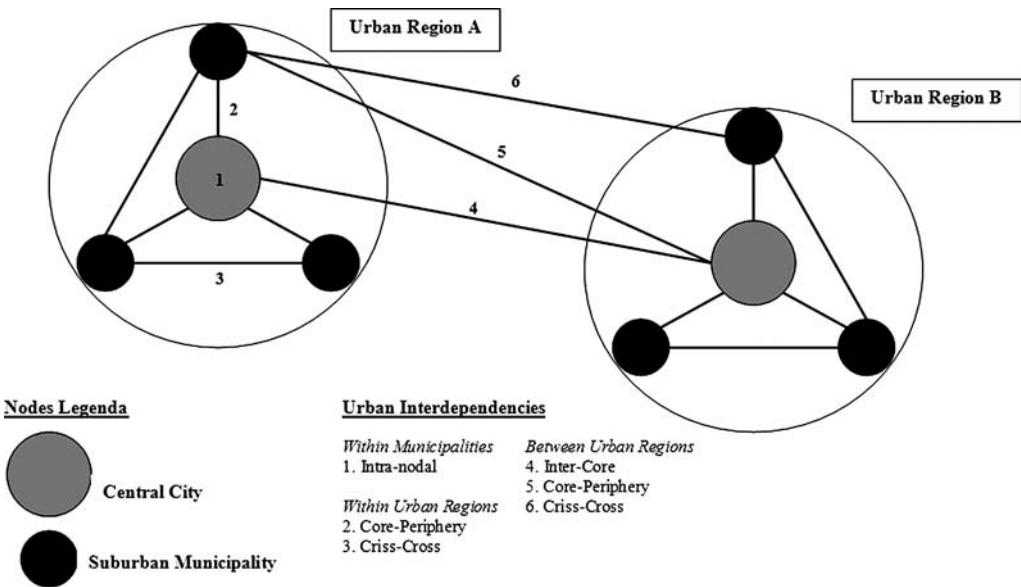


Figure 4. Classification of different types of urban interdependencies.

C. Economic relations between urban (sub-) regions

- 4. Intercore interdependencies between the central cities of Amsterdam, Rotterdam, The Hague and Utrecht.
- 5. Core-periphery interdependencies between the central cities and one of the non-‘own’ suburban municipalities.
- 6. Interperiphery or criss-cross interdependencies between suburban municipalities that are not situated in the same urban region.

The different types of urban interdependencies are displayed in Figure 4. Recall that, if the Randstad can be characterised as a spatially integrated urban system (from an economic point of view), the regional network structure of interfirm relations is solely determined by the masses of the municipalities and the physical distances between them; spatial context does not play any role and there is no difference in the relative strength of the different types of interdependencies. The

advantage of this testing system is that it is independent of the scale of regions and cities and can be applied in other settings (for a comparable test on urban network formation in the Greater South East, in the UK, see de Goei *et al.*, 2010).

4.2 Empirical Results

Table 2 presents the estimates from the zero-inflated negative binomial regression model for the economic interaction intensity between municipalities in the Randstad region. Overall, the significance values of the likelihood ratio test of overdispersion (α) and the Vuong-statistic indicate that the zero-inflated negative binomial model fits the data best, which justifies the choice of this model. As indicated earlier, the zero-inflated negative binomial model consists of two components. The negative binomial component corresponds to the estimates for those pairs of municipalities with non-strictly zero counts. The zero-inflated or binary component relates to the logit regression predicting whether

Table 2. Zero-inflated negative binomial models on economic interaction in the Randstad

	<i>Model 1</i> (<i>N</i> = 2415)	<i>Model 2</i> (<i>N</i> = 2415)	<i>Model 3</i> (<i>N</i> = 2346)	<i>Model 4</i> (<i>N</i> = 2346)
<i>Negative binomial part</i>				
Intercept	−4.86 (22.2)***	−4.66 (21.6)***	−6.18 (23.8)***	−5.79 (14.1)***
Mass (ln)	0.91 (36.0)***	0.88 (37.2)***	1.02 (41.7)***	0.94 (23.6)***
Distance (ln)	−0.85 (23.5)***	−0.63 (9.66)***	−0.58 (9.23)***	−0.58 (8.67)***
Intramunicipal		•	—	—
<i>Other within region</i>		−0.58 (4.35)***	•	
Core–periphery				0.46 (2.88)***
Criss-cross				0.24 (1.48)
<i>Between regions</i>		−0.75 (3.81)***	−0.28 (2.72)***	
Intercore				0.52 (2.47)***
Core–periphery				0.13 (0.91)
Criss-cross				•
Overdispersion (α)	−7.95***	−8.48***	−6.31***	−6.38***
<i>Zero-inflated part</i>				
Intercept	−0.54 (0.34)	−11.6 (0.02)	−10.6 (0.03)	2.59 (1.32)
Mass (ln)	−0.62 (4.25)***	−0.67 (5.42)***	−0.61 (4.21)***	−0.55 (2.83)***
Distance (ln)	1.03 (2.70)***	0.70 (0.19)	0.16 (0.36)	0.17 (0.42)
Intramunicipal		•	—	—
<i>Other within region</i>		11.8 (0.02)	•	
Core–periphery				−15.2 (0.02)
Criss-cross				−2.31 (1.19)
<i>Between regions</i>		15.6 (0.02)	13.2 (0.04)	
Intercore				−31.0 (0.00)
Core–periphery				−0.62 (0.76)
Criss-cross				•
Vuong-statistic	1.45*	2.39***	1.98**	1.83**
Log likelihood	−1575	−1556	−1400	−1394
McFadden's	0.360	0.366	0.349	0.349
Adjusted R^2				
AIC	1.310	1.297	1.201	1.201

Notes: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. Absolute z-values between brackets. • = benchmark. For the overdispersion and Vuong-statistics, the values of the z-test are displayed.

an observation belongs to the 'always zero' or 'not always zero' group. Examining the zero-inflated component of the model, it can be observed that in three of the four models presented, only a low joint mass significantly increases the probability of belonging to the 'always zero' group. On average, an increase

of 1 per cent in the joint mass of the two municipalities decreases the likelihood of belonging to the non-strictly zero group by 0.5 per cent, holding all other variables constant. This is in line with the finding that the largest share of non-interaction is between many small localities, where the critical mass

to generate flows is lacking. In the remainder of our discussion of the empirical results, we focus on the negative binomial component of the regression model.

Model 1 in Table 2 represents the null or baseline model and only includes variables of the joint mass of the originating and destination municipalities and the physical distance between them. Recall that, if the Randstad is functioning as one urban network, the joint mass and the physical distance between municipalities should solely determine the interaction intensity between municipalities in the Randstad. The other types of spatial interdependency should not play any role. As expected, economic mass has a marked direct correlation with the flow frequency between the municipalities, while distance has a marked inverse correlation. These values can be interpreted as elasticities: when physical distance increases by 1 per cent, the interaction intensity of firm relations is predicted to decrease by 0.85 per cent. Similarly, an increase in the joint mass of the two municipalities by 1 per cent increases the predicted interaction intensity of firm relations by 0.91 per cent.

Model 2 in Table 2 tests the first condition for spatial integration, which states that the intranodal interdependency should not be stronger than interdependencies between municipalities within the Randstad region. In other words, the degree of spatial interaction between firms that remains within municipalities should not exceed the degree of interaction between municipalities. Controlling for mass and physical distance between municipalities in our model, it appears, however, that this condition is not met. The intraurban interdependency is significantly stronger than interdependencies between municipalities situated in the same region as well as interdependencies between other municipalities. More specifically, the predicted interaction intensity within municipalities is over 75 per cent higher than that between municipalities within the same region. Likewise, *vis-à-vis*

interdependencies between municipalities not situated in the same region, this figure is approximately 110 per cent. (These marginal elasticities can be obtained by taking the natural exponent of the parameter estimates.)

In Model 3 in Table 2, the second condition for spatial integration is tested, which states that the interdependencies between cities within one of the urban (sub-)regions in the Randstad should not be stronger than the interdependencies between cities across these (sub-)regions. This condition also is not met. The interaction intensity between municipalities within the same region is predicted to be significantly higher than the interaction intensity between municipalities across urban (sub-)regions. Holding mass and physical distance constant, the predicted flow between municipalities within the same urban region is over 30 per cent larger compared with the intermunicipal flows that exceed the level of the urban region.

In Model 4 in Table 2, we test the third condition for spatial integration, which states that there should be no observable hierarchy in the different types of intermunicipal interdependencies present. Taking interregional criss-cross interdependencies as a reference category (as it is conceptually the weakest type of urban interdependency), we are able to compare the relative strength of the different types of urban interdependencies, controlling for mass and distance. We conclude whether the interaction intensity significantly differed between the different types by using a Wald test to check the equality of coefficients. This way, we are able to compile a ranking list, in which the different types of urban interdependencies were ranked according to relative strength. From this analysis, we conclude that the third condition for spatial integration is also not met. Intercore interdependencies between the four largest cities (Amsterdam, Rotterdam, The Hague and Utrecht) and intraregional core-periphery interdependencies are the strongest types of urban interdependencies. Interregional core-periphery interdependencies and intraregional

criss-cross interdependencies are the weakest types of relationship. As none of the three conditions for spatial integration is formally met in our tests, it can be concluded that the Randstad does not (yet) function as a spatially integrated cluster. Relations between firms appear to be both regionalised within the agglomeration level of the four (sub-)regions and interregionalised between the four larger cities.³

5. The Functional Economic Integration of Randstad Holland

5.1 Measuring Functional Economic Integration and Complementarities

Although spatial integration is one prerequisite for urban complementarities, functional integration or the existence of a spatial division of labour is the other one. Functional economic integration presupposes a differentiation between cities within the Randstad in terms of economic specialisations and functions, while at the same time a large degree of integration (the use of each others' specialisations) reflects the existence and use of an aggregated regional production system. In this section, we test whether functional integration is present in the Dutch Randstad region.

Since we want simultaneously to test for spatial integration (as defined in the previous section) and functional integration, we extend our gravity models on spatial interaction between firms with an indicator for functional integration. When firms in municipalities i and j have many mutual relations, this can be rooted in the fact that the local production structure in i is characterised by other—complementary—specialisations in j . For example, a city specialising in financial services provides these services to (firms in) a city specialising in labour-intensive industry and vice versa (Lambregts *et al.*, 2006). In theory, the business services specialisation in Amsterdam should lead to a large number of companies in other regions making use of these services, not least because the services

are underrepresented in their own immediate region. Companies outside Amsterdam are then able to specialise in another type of activity, which will then attract trading relationships from yet other regions. In this scenario, each region benefits from each other's specialisation and will still have marked interactions with each other. Hence, cities do not have to be specialised in all possible sectors, but can benefit from specialisations elsewhere in the urban network (Meijers, 2005).

We speak of the presence of urban complementarities if functional integration coincides with or complements spatial integration. We measure the relative specialisation of municipalities by means of location quotients in 27 different basic sectors.⁴ Basic sectors are not dependent on the direct location of consumers for their choice of location and hence firms in these sectors may profit from complementary specialisations in (nearby) cities.⁵ The location quotients are defined as:

$$LQ_{i,j} = \frac{E_{i,j} / \sum_j E_{i,j}}{\sum_i E_{i,j} / \sum_{i,j} E_{i,j}} \quad (5)$$

in which, E represents the number of firms or employment in region i and sector j .

A score larger than one indicates that the sector is relatively overrepresented in the region when compared with the Randstad on average. A score lower than 1 indicates the relative under-representation of a sector in a municipality. The location quotient is expressed in terms of employment and in terms of number of firms. The difference between the two reflects the average firm size in sectors in locations. It is not clear *a priori* whether specialisations of the number of firms or specialisations in employment are related to functional urban complementarities. We therefore weigh both equally in our indicator for functional integration (FI), which for each economic interaction between origin and destination municipalities ($n = 2346$) is defined as⁶

$$FI_{OD} = \sum_j |LQo_j^E - LQd_j^E| + \sum_j |LQo_j^F - LQd_j^F| \quad (6)$$

in which, LQo represents location quotients of origin municipalities; LQd represents location quotients of destination municipalities; E stands for employment; F for the number of firms; and j for the 27 sectors distinguished.

This indicator measures and aggregates all possible functional complementarities between the 27 sectors and between the 69 municipalities in the Randstad region. Functional complementarities at first are per municipality, expressed as a weighted average of the difference in specialisations between all potential origin and destination combinations. These differences are then matched with the actual origin and destination municipalities in the measured economic interactions between municipalities.

5.2 Empirical Results

Table 3 presents the estimates from the zero-inflated negative binomial regression model for the economic interaction intensity between municipalities in the Randstad region, complemented with the indicator for functional integration.⁷ Overall, the likelihood ratio test of overdispersion (α) and the Vuong-statistic again indicate that the zero-inflated negative binomial model fits the data best. The zero-inflated component of the model reveals similar results as those presented in Table 2. Model 5 in Table 3 represents the null or baseline model when functional integration is included along with the variables of the joint mass of the originating and destination municipalities and the physical distance between them. The results indicate that there is no significant effect of functional integration on flow frequency between cities.

Model 6 in Table 3 shows a negative significant interaction effect of the functional integration indicator with the mass indicator of municipalities in origin and destination

municipalities. This can be interpreted as weak evidence of functional integration within a group of relative small cities and municipalities in the Randstad. As these cities are too small to house all economic functions individually, functional complementarities with nearby municipalities containing other specialisations occur. Larger cities have all the specialised resources needed for firm networks available within their boundaries. Models 7 and 8 introduce spatial and functional integration variables simultaneously in the model. While all spatial interaction variables show similar relations with firm interactions as presented earlier in Models 2–4 in Table 2, the introduction of the indicator for functional integration is only slightly significant, again only for smaller municipalities.⁸

6. Conclusions

In this article, we have examined whether the Randstad can be regarded as an urban network and an integrated economic entity. There is a need for this, since a burgeoning literature suggests that the polycentric region as a spatial economic concept replaces the hierarchical, central node concept. Camagni and Capello (2004) argue that polycentric urban regions only comprise a new paradigm in spatial sciences when: the exact meaning is defined; the theoretical economic rationale is justified; and, the novel features of its empirical content are clearly identified and distinguished from more traditional spatial facts and processes that can be interpreted through existing spatial paradigms. The concept of spatial networks in the empirical literature is merely used as a substitute for ‘interaction’ (exchange of goods, services, information and contacts among places and nodes), for which traditional paradigms of spatial interaction can be utilised. For the concept of urban networks

Table 3. Zero-inflated negative binomial models on economic interaction and functional integration in the Randstad ($N = 2346$)

	<i>Model 5</i> ($N = 2415$)	<i>Model 6</i> ($N = 2415$)	<i>Model 7</i> ($N = 2346$)	<i>Model 8</i> ($N = 2346$)
<i>Negative binomial part</i>				
<i>Intercept</i>	-5.47 (11.1)***	-10.8 (5.65)***	-5.17 (8.84)***	-9.68 (3.98)***
<i>Mass (ln)</i>	1.01 (32.8)***	1.62 (7.64)***	0.92 (21.4)***	1.42 (5.43)***
<i>Distance (ln)</i>	-0.72 (15.6)***	0.71 (16.1)***	0.57 (8.37)***	-0.54 (7.97)***
<i>Intramunicipal</i>	—	—	—	—
<i>Other within region</i>				
Core-periphery	—	—	0.47 (3.00)***	0.48 (2.96)***
Criss-cross	—	—	0.24 (1.53)	0.30 (1.76)*
<i>Between regions</i>				
Intercore	—	—	0.53 (2.52)**	0.36 (1.64)*
Core-periphery	—	—	0.14 (0.94)	0.14 (0.94)
Criss-cross	—	—	•	•
<i>Functional integration</i>				
Δ Relative specialisation	-0.10 (1.06)	1.50 (2.64)***	-0.14 (1.53)	1.15 (1.70)*
Δ Relative specialisation* mass		-0.18 (2.89)***	—	-0.15 (1.95)*
Overdispersion (α)	-6.60***	-6.58***	-6.15***	-5.38***
<i>Zero-inflated part</i>				
<i>Intercept</i>	0.51 (0.13)	3.00 (0.38)	4.77 (3.12)***	7.61 (0.84)
<i>Mass (ln)</i>	-0.68 (3.02)***	-1.34 (1.36)	-0.63 (3.00)***	-1.46 (1.33)
<i>Distance (ln)</i>	1.37 (2.27)**	1.01 (2.52)	-0.19 (0.46)	-13.5 (0.04)
<i>Intramunicipal</i>	—	—	—	—
<i>Other within region</i>				
Core-periphery	—	—	-15.3 (0.02)	1.22 (1.24)
Criss-cross	—	—	-2.21 (1.41)	-26.5 (0.00)
<i>Between regions</i>				
Intercore	—	—	-29.7 (0.00)	-0.53 (0.74)
Core-periphery	—	—	-0.76 (0.88)	0.30 (0.79)
Criss-cross	—	—	•	•
<i>Functional integration</i>				
Δ Relative specialisation	-0.61 (0.72)	-0.87 (0.41)	-0.47 (0.91)	-1.38 (0.58)
Δ Relative specialisation* mass	—	0.19 (0.47)	—	0.24 (0.80)
Vuong-statistic	1.67**	1.93**	1.90**	2.13**
Log likelihood	-1409	-1405	-1393	-1388
McFadden's Adjusted R^2	0.343	0.345	0.348	0.349
AIC	1.211	1.207	1.202	1.200

Notes: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. Absolute z-values between brackets. • = benchmark. For the overdispersion- and Vuong-statistics, the values of the z-test are displayed.

to be considered different from that of cities or urban agglomerations, the probability of interfirm interaction should be independent of distance, the size of nodes and hierarchical relations between nodes. Functionally, the contention for different cities within an urban network to be complementary to each other requires that cities not only be specialised in different industries but at the same time also be showing a marked degree of spatial interaction and hence integration. To date, there has been little empirical research into the spatial and functional economic cohesion of urban networks, mainly because of the lack of data on interfirm relations.

As the Randstad combines Amsterdam (cultural capital), The Hague (political capital), Rotterdam (gateway) and Utrecht (central national position), together with numerous smaller cities and a highly skilled labour force, the region is traditionally regarded as the European showcase of regional polycentricity (Lambregts *et al.*, 2006). Relationships between the cities date back centuries. No earlier empirical research tests whether this image is justified for present-day economic (interfirm) relations. We tested for the spatial and functional integration of the Randstad and the presence of urban complementarities, relying on recently gathered data that derive from a large survey of 1676 firms in the region. We formulated a set of three conditions for *spatial* integration within the Randstad region. None of these three conditions for spatial integration was met. First, we found that intraurban economic interdependencies are stronger than interdependencies between cities within the Randstad. Secondly, interdependencies between cities within one of the urban (sub-)regions in the Randstad defined by the largest cities were stronger than interdependencies between cities across these (sub-)regions. And, thirdly, an observable hierarchy in the different types of intermunicipal interdependencies in the Randstad is present in which central place relations prevail. Therefore, it can be concluded that the Randstad does not

(yet) function as a spatially integrated network of cities. We then introduced an indicator for *functional* integration based on the differences in economic specialisations of origin and destination municipalities that are included in the firm relational network data. We found no clear evidence for functional integration over municipalities in the Randstad. Urban complementarities in the Randstad defined by spatial and functional integration between municipalities are thus currently not present. The largest cities—Amsterdam, Rotterdam, The Hague and Utrecht—are all mature nodes within their own functional economic region. Weak evidence for urban complementarities is only found for the smallest municipalities that cannot accommodate all economic functions within their boundaries.

The economic (interfirm) dimension of urban networks has not been put systematically to an empirical test. Our research results question the many policy attempts to create and sustain polycentric economic development trajectories in Europe. When economic complementarities based on interfirm linkages do not exceed the (monocentric) city-region in the ‘archetypical polycentric’ Randstad, it is doubtful that policy aiming at higher-level interaction can be justified, outside wishful thinking. Our results suggest that a focus on the daily urban systems (DUS) of the four largest cities fits the spatial-economic reality better. Recall that 50 per cent of the interfirm relations are with (inter)national regions outside the Randstad, which causes spatial and functional dependencies to transcend the DUS. However, the scale of the polycentric urban region appears to be skipped in the so-called local–global economic development trajectories (Bathelt *et al.*, 2004). More comparable research in other regions should be carried out in order to confirm or confront this outcome. The testing system introduced is independent of the scale of regions and cities and can therefore be applied to this end. We think that this testing system, although strictly formulated, is

advantageous for the testing of hypotheses. Less strict testing will, in our opinion, leave much room for speculation on the potential (policy) usage of the urban network paradigm. It is also important repeatedly to measure inter-firm relationships. Although commuting and shopping relations do not suggest a profound development towards more interregional network formation in the Randstad for the past 20 years (RPB, 2006), interfirm relations may develop in that direction over time.

Notes

1. Over the past 20 years, the average distance travelled by consumers for shopping activities has not grown significantly in the Netherlands, either (RPB, 2006).
2. A more extensive discussion of zero-inflated estimation in relation to spatial interaction models can be found in Burger *et al.* (2009).
3. Because agglomeration economies differ across industries (Rosenthal and Strange, 2004), we tested for the robustness of the degree of spatial integration in interaction models for manufacturing, distribution and business service activities separately. Since industrial activities are to a considerable extent also located outside the Randstad (van Oort, 2004), the model of business services resembles the (total) model presented to a large degree. Because we are interested in functional integration by means of detailed *cross-sectoral* interdependencies (see section 5), we did not explore the three sector-specific interaction models any further.
4. These 27 sectors are (in brackets sectoral Dutch sectoral SBI-codes): the food and beverage industry (15), the tobacco industry (16), the textile industry (17), the clothing industry (18), the leather and leather goods industry (19), the timber industry (20), the paper industry (21), the oil and coal industry (23), the chemical products industry (24), the rubber industry (25), the glass and ceramics industry (26), the primary metal industry (27), the metal products industry (28), the machine industry (29), the computer industry (30), the electronics industry (31), the audio and telecom industry (32), the medical equipment and instruments industry (33), the car industry (34), the transport equipment industry (35), the furniture industry (36), wholesale (51), the publishing industry (22), telecommunication services (64), computer services and consultancies (72), the research and development and knowledge institutions (73) and the remaining business services (74).
5. Non-basic sectors are retail, primary education and schools, local public services like police, fire departments and healthcare, and local and regional governments.
6. Analyses with location quotients for employment and the number of firms present in municipalities separately give similar results to those presented. Also, other (aggregated) definitions of sectors do not change the outcomes.
7. Correlations between explanatory variables in the models are not higher than 0.4, meaning that there is little risk of multicollinearity problems.
8. For both models 6 and 8, we find no significant interaction effect of functional integration and distance. For this reason, this interaction term is omitted from our final models.

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